

Computational Stylometric Analysis and Authorship Attribution: An Exploratory Comparative Study of the Literary Styles of Mark Twain and Charles Dickens

by Constantinos Panayi

Postgraduate Student, MSc Digital Humanities

National and Kapodistrian University of Athens, University of Cyprus, ATHENA Research Centre

Abstract—This study presents an exploratory computational stylometric analysis of twenty literary works authored by Mark Twain and Charles Dickens. Within a Digital Humanities (DH) framework, the investigation combines stylometric feature extraction, dimensionality reduction, and machine learning classification to determine whether quantifiable linguistic patterns can effectively distinguish between the two authors. Four distinct dimensions are examined, namely lexical diversity, syntactic structure, grammatical composition, and affective expression. Pearson correlation analysis, Principal Component Analysis (PCA), and Binomial Logistic Regression are employed to identify stylistic structures and evaluate authorship attribution performance. The findings indicate that vocabulary diversity, sentence length, and sentiment-related features provide the strongest discriminatory information, whereas grammatical distributions contribute comparatively little. Subsequently, PCA revealed meaningful author-related separation, while Logistic Regression achieved high classification performance. Collectively, the results demonstrate the utility of quantitative stylometry in literary studies. The present study was conducted as the final project for the *Data Analysis for Humanities with Python* course of the MSc Digital Humanities programme, jointly offered by the *National and Kapodistrian University of Athens*, the *University of Cyprus*, and the *ATHENA Research Centre*.

Index Terms—Digital Humanities, Computational Stylometry, Authorship Attribution, Natural Language Processing, Sentiment Analysis, Statistical Analysis, Machine Learning, Principal Component Analysis, Logistic Regression

I. INTRODUCTION

The present study aims to develop and present an exploratory computational stylometric framework for the comparative analysis and authorship attribution of literary texts written by Twain and Dickens. By combining statistical text analysis, stylometric feature extraction, dimensionality reduction techniques, and machine learning classification, the study examines whether quantifiable linguistic patterns can effectively distinguish between the writing styles of the two authors.

From a methodological perspective, the study draws upon computational stylometry, a branch of DH concerned with the quantitative analysis of literary style through measurable textual features. Unlike traditional stylistic analysis, which primarily relies on qualitative interpretation and close reading, computational approaches employ statistical and algorithmic techniques to identify recurring patterns in language use. Thus, such methods have become particularly important in authorship attribution, where stylistic evidence is used to determine or verify the authorship of a text. [1]

In broad terms, stylometric analysis may be conducted across several complementary dimensions. More specifically:

- **Lexical characteristics** focus on vocabulary usage and lexical diversity, providing insights into the richness and variation of an author's word choices.
- **Syntactic properties** examine structural aspects of writing, such as sentence length, sentence complexity, and the use of passive constructions.
- **Grammatical patterns** investigate the distribution of linguistic categories, including Part-of-Speech (POS) classes, which often reflect individual writing preferences.
- **Affective features** explore the emotional and attitudinal characteristics of language through sentiment analysis, offering an additional perspective on authorial expression.

Collectively, these dimensions provide a multidimensional representation of literary style and authorial behaviour.

A. Research Aims

Within this context, the primary objective of the study is to examine whether stylometric features can effectively differentiate between the literary styles of Twain and Dickens and, consequently, support automated authorship attribution.

More specifically, the study seeks to address the following research questions:

- Are there measurable lexical differences between the works of the two authors?
- Are there measurable syntactic differences between their writing styles?
- Do the authors exhibit distinct grammatical patterns?
- Are there detectable differences in the affective characteristics of their texts?
- Do stylometric features produce meaningful author-based clusters?
- Can machine learning methods accurately classify texts according to their author?

Ultimately, by addressing these questions, the study aims to evaluate both the interpretative value of stylometric features for literary analysis and their effectiveness for computational authorship attribution. This comprehensive approach is synthesised in Figure 1, which illustrates the complete analytical workflow adopted throughout the investigation.

II. DATA MANAGEMENT

A. Corpus Overview

The corpus employed in the present study consists of twenty literary works authored by Twain (1835–1910) and Dickens (1812–1870), two of the most influential novelists of the nineteenth century. Twain was an American novelist, essayist, and humourist whose works are widely regarded as foundational texts of American literature, while Dickens was an English novelist and social critic whose writings played a central role in the development of Victorian literature. [2], [3]

These authors were selected for examination due to their literary significance, extensive textual output, and distinct

TABLE I: Corpus Description

Author	Work Title
Mark Twain	The Innocents Abroad (1869)
	Roughing It (1872)
	The Adventures of Tom Sawyer (1876)
	The Prince and the Pauper (1881)
	Life on the Mississippi (1883)
	Adventures of Huckleberry Finn (1884)
	A Connecticut Yankee in King Arthur's Court (1889)
	The Tragedy of Pudd'nhead Wilson (1894)
	Personal Recollections of Joan of Arc (1896)
	The Mysterious Stranger (1916)
Charles Dickens	The Pickwick Papers (1837)
	Oliver Twist (1838)
	Nicholas Nickleby (1839)
	Barnaby Rudge (1841)
	David Copperfield (1850)
	Bleak House (1853)
	Hard Times: For These Times (1854)
	Little Dorrit (1857)
	A Tale of Two Cities (1859)
	Great Expectations (1861)

cultural and stylistic backgrounds. Additionally, both novelists are widely recognised for their narrative prose and enduring influence on English-language literature, making them particularly suitable for a comparative stylometric investigation.

Finally, the resulting corpus comprises ten works from each author, forming a balanced literary dataset that minimises potential biases associated with unequal author representation (Table I). All texts were collected from the *Project Gutenberg* digital library [4] and converted into a consistent encoding format, ensuring compatibility with the subsequent Natural Language Processing (NLP) pipeline. [5]

B. Exploratory Corpus Analysis and Quality Assessment

Prior to feature extraction and modelling, an initial exploratory assessment of the corpus was conducted to establish a general understanding of its structural characteristics. At this stage, attention was limited to high-level descriptive properties of the texts, including *total word count*, *vocabulary size*, *total number of sentences*, and *average sentence length*.

These measures provide a preliminary overview of corpus variability and facilitate the identification of potential irregularities that might affect following analyses. Descriptive statistics and visual exploration were therefore employed to examine the distribution of these characteristics across the collected works (Figure 2).

From an interpretative perspective, the boxplots reveal several notable patterns. Dickens' works generally contain more words than those of Twain, reflecting the

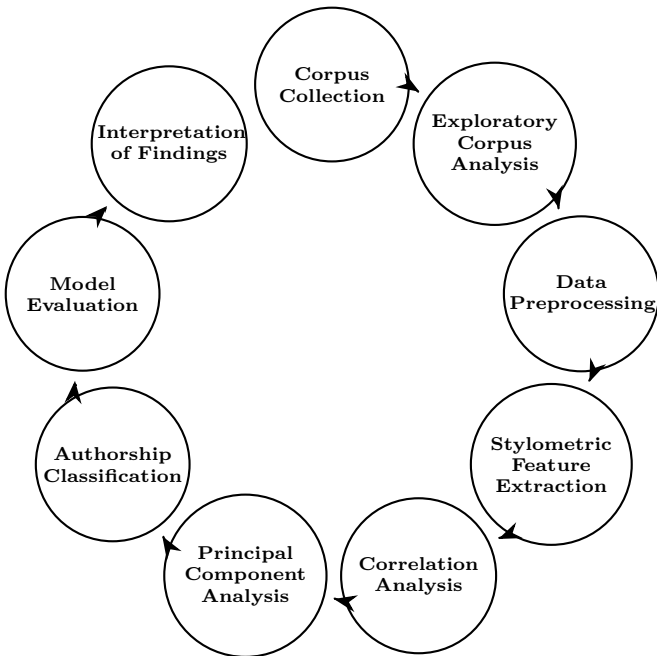


Fig. 1: Analytical workflow of the study.

Corpus-level characteristic distributions across authors

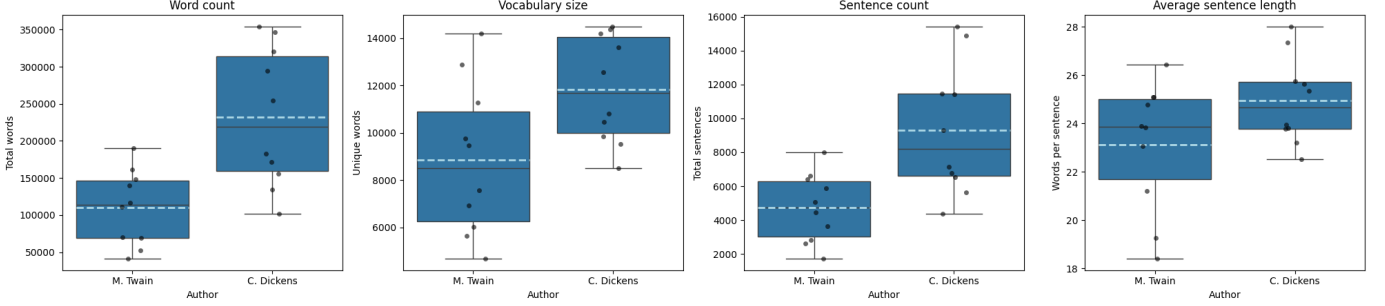


Fig. 2: Corpus-level characteristic distributions across authors.

larger scale of many Victorian novels. Similarly, Dickens exhibits higher vocabulary sizes overall, although Twain’s texts display substantial lexical variation despite their comparatively shorter length. Sentence counts also tend to be higher for Dickens, indicating longer works with a greater number of textual units. In contrast, average sentence length shows only minor differences between the two authors, with Dickens exhibiting a slightly higher central tendency and greater syntactic consistency.

Overall, the exploratory assessment suggests that Dickens’ works are generally larger in scale, whereas Twain exhibits comparable levels of lexical diversity within shorter texts. At the same time, the considerable overlap observed across all four characteristics indicates that corpus-level measures alone are insufficient for reliable authorship discrimination.

C. Data Preprocessing

Following corpus validation, a preprocessing pipeline was implemented to prepare the texts for stylometric analysis. Three complementary corpus representations were created in order to support different stages of the analytical workflow.

- **Raw Corpus:** the original texts were preserved without modification and were primarily used for *sentence-level analyses*, *syntactic feature extraction*, and *sentiment analysis*.
- **Sentence Corpus:** texts were segmented into individual sentences to facilitate sentence-based calculations, including *average sentence length*.
- **Cleaned Corpus:** texts were tokenised and subsequently processed through a series of cleaning procedures to support *lexical* and *grammatical analyses*.

The cleaning process included *lowercasing*, *punctuation removal*, and *stopword removal*. These procedures were applied in order to reduce noise and improve the reliability of lexical measurements while retaining the stylistically meaningful vocabulary employed by each author.

Importantly, *lemmatisation* was intentionally avoided. Since the objective of the study is to investigate stylistic variation rather than semantic equivalence, preserving

the original lexical forms allows the analysis to capture potentially informative differences in word choice and grammatical expression. Consequently, the full stylistic vocabulary of each author was retained throughout the feature extraction process.

III. METHODOLOGY

A. Stylometric Feature Extraction

As outlined in the Introduction, literary style was examined through four complementary analytical dimensions. Each dimension captures different aspects of authorial expression and collectively provides a multidimensional representation of writing style. Based on these stylistic axes, a total of fourteen stylometric features were extracted from each text and subsequently used for the statistical and machine learning analyses.

1) *Lexical Features:* Lexical features describe the richness, diversity, and variation of an author’s vocabulary. In the present study, lexical style was quantified using three complementary measures, where higher values indicate greater vocabulary variation. [6]

a) **Type-Token Ratio (TTR):** TTR measures lexical diversity by comparing the number of unique words to the total number of words within a text.

$$TTR = \frac{N_{types}}{N_{tokens}} \quad (1)$$

where N_{types} represents the total number of unique words and N_{tokens} the total number of words.

This metric is simple, intuitive, and computationally efficient; however, it is highly sensitive to text length, limiting direct comparisons across texts of substantially different sizes.

b) **Measure of Textual Lexical Diversity (MTLD):** MTLD measures lexical diversity by estimating the average length of word sequences required to reach a predefined TTR threshold (0.72).

$$MTLD = \frac{N_{tokens}}{N_{sequences}} \quad (2)$$

where N_{tokens} represents the total number of tokens and $N_{sequences}$ the number of completed word sequences.

This measure provides a more robust estimate of lexical diversity by reducing the influence of length. Nevertheless, its computation is more complex and its interpretation is generally less intuitive.

c) Hypergeometric Distribution Diversity (HD-D): HD-D estimates lexical diversity using probability-based random sampling of word types.

$$HD-D = \sum P(word_i) \quad (3)$$

where $P(word_i)$ denotes the hypergeometric probability associated with word type i .

This metric offers a relatively stable estimate of lexical diversity across texts of different lengths. However, the method is computationally more demanding and less transparent than other simpler measures.

2) Syntactic Features: Syntactic features capture structural characteristics of written language and provide insights into sentence construction.

a) Average Sentence Length (ASL): ASL measures the average number of words per sentence and serves as an indicator of syntactic complexity.

$$ASL = \frac{N_{words}}{N_{sentences}} \quad (4)$$

where N_{words} represents the total number of words and $N_{sentences}$ the total number of sentences.

This method is straightforward to calculate and easily interpretable. However, it does not capture internal grammatical complexity beyond length.

b) Passive Voice Rate (PVR): PVR measures the proportion of passive constructions within a text and is often associated with formality and narrative distance.

Passive constructions were identified using grammatical patterns combining forms of *be*, *have*, *get*, and modal verbs followed by a past participle.

$$PVR = \frac{N_{passive}}{N_{sentences}} \quad (5)$$

where $N_{passive}$ represents the number of passive constructions and $N_{sentences}$ the total number of sentences.

This measure captures stylistic preferences related to agency and sentence focus. Nevertheless, rule-based detection may occasionally miss passive constructions or generate false positive identifications.

3) Grammatical Features: Grammatical style was represented through **POS distributions**. POS analysis quantifies the relative frequency of grammatical categories within a text and provides insight into an author's linguistic preferences. The texts were tagged using the Natural Language

Toolkit (NLTK), and five grammatical categories were retained: *nouns*, *verbs*, *adjectives*, *adverbs*, and *pronouns*.

For each grammatical class, the relative proportion was calculated as:

$$POS \text{ Percentage} = \frac{N_{POSClass}}{N_{tokens}} \times 100 \quad (6)$$

where $N_{POSClass}$ represents the frequency of a specific grammatical class and N_{tokens} the total number of tokens.

This metric provides interpretable indicators of grammatical style. However, broad grammatical categories may obscure more subtle linguistic distinctions present within individual texts.

4) Affective Features: Affective style was examined using sentiment analysis. Specifically, the **Valence Aware Dictionary and Sentiment Reasoner (VADER)** was employed to quantify the emotional characteristics of each text. [7] Four sentiment scores were subsequently extracted: *negative*, *neutral*, *positive*, and *compound*. The first three quantify the relative intensity of each sentiment within a text, whereas the latter represents a weighted measure capturing the overall sentiment polarity.

Finally, VADER was selected because it incorporates sentiment valence together with contextual cues such as punctuation, capitalisation and negation. However, dictionary-based approaches may struggle with literary language, irony, and context-dependent meanings.

B. Modelling Approach

Following feature extraction, a series of statistical and machine learning techniques were employed to analyse the resulting stylometric feature space. Correlation analysis was initially conducted to identify linear relationships among variables. Subsequently, PCA was applied to detect underlying patterns and potential author-based clusters. Ultimately, Logistic Regression was implemented to perform authorship classification and evaluate the predictive ability of the extracted stylometric features.

1) Statistical Correlation Analysis: First, a correlation analysis was performed to examine the relationships among the extracted stylometric features and identify potential multicollinearity. Specifically, the **Pearson Correlation Coefficient (r)** was employed to quantify the strength and direction of linear associations between pairs of variables.

The Pearson Correlation Coefficient is defined as:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (7)$$

where x_i and y_i denote paired observations, while $\bar{x}$ and $\bar{y}$ represent their corresponding sample means.

Pearson correlation provides a simple and efficient measure of linear dependence between variables. The resulting correlation matrix was visualised through a heatmap to

facilitate the detection of highly correlated feature pairs. However, this linear model may overlook more complex associations.

2) *Dimensionality Reduction*: **PCA** was employed to reduce the dimensionality of the stylometric feature space while preserving most of its variability. [8] The method is based on the covariance structure of the data and transforms the original standardised variables into orthogonal components defined by *eigenvectors*, which indicate directions of maximum variance, while the corresponding *eigenvalues* quantify the variance captured by each component.

a) *Explained Variance Ratio (EVR)*: The contribution of each component was evaluated using the **EVR**, which represents the proportion of total variance accounted for by a given component.

$$EVR_i = \frac{\lambda_i}{\sum_{j=1}^p \lambda_j} \quad (8)$$

where λ_i denotes the eigenvalue of the i -th principal component, λ_j the eigenvalue of component j , and p the total number of principal components.

b) *Principal Component Loadings*: To facilitate interpretation, **Principal Component Loadings** were examined. Loadings quantify the importance of each original feature to a component and indicate the variables underlying the observed patterns.

$$Loading_{ij} = v_{ij} \sqrt{\lambda_j} \quad (9)$$

where v_{ij} denotes the eigenvector coefficient of feature i on component j , and λ_j the corresponding eigenvalue.

Consequently, PCA provides an effective framework for uncovering latent stylistic patterns and visualising high-dimensional data, while component loadings highlight the features contributing most strongly to each resulting component. However, the method assumes linear relationships among variables, while the resulting axes may not always be straightforward to interpret in stylistic terms.

3) *Classification*: To evaluate the predictive ability of the extracted stylometric features, a **Binomial Logistic Regression** classifier was implemented. Logistic Regression estimates the probability that a text belongs to a particular author by modelling the linear relationship between the predictor variables and a binary outcome. [9]

$$P(Y = 1) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_1 + \dots + \beta_n X_n)}} \quad (10)$$

where $P(Y = 1)$ denotes the probability of belonging to a given class, X_i the predictor variables, and β_i the corresponding model coefficients.

a) *Stratified K-Fold Cross Validation*: To assess model robustness, **Stratified K-Fold Cross Validation** was employed. The dataset was partitioned into five equally sized folds while preserving the original class distribution. Each fold was used once for testing and four times for training, thereby providing a more reliable estimate of model performance and reducing the risk of overfitting.

b) *Confusion Matrix*: Model predictions were summarised using a **Confusion Matrix**, which compares predicted and observed class labels and provides the basis for the calculation of classification performance metrics.

c) *Evaluation Metrics*: Classification performance was evaluated using Precision, Recall, Accuracy, and F1-score.

$$Precision = \frac{TP}{TP + FP} \quad (11)$$

$$Recall = \frac{TP}{TP + FN} \quad (12)$$

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (13)$$

$$F1 = 2 \cdot \frac{Precision \times Recall}{Precision + Recall} \quad (14)$$

where TP denotes the number of True Positives, TN the True Negatives, FP the False Positives, and FN the corresponding False Negatives.

Accordingly, each metric captures a different aspect of classification performance:

- **Precision**: Proportion of predicted positive instances correctly classified.
- **Recall**: Proportion of actual positive instances correctly identified.
- **Accuracy**: Proportion of correctly classified instances.
- **F1-score**: Harmonic mean of Precision and Recall.

d) *Receiver Operating Characteristic (ROC) curve and Area Under the Curve (AUC)*: Model discrimination was further assessed using the **ROC curve**, which visualises the trade-off between the True Positive Rate (TPR) and the False Positive Rate (FPR) across different classification thresholds.

$$TPR = \frac{TP}{TP + FN} \quad (15)$$

$$FPR = \frac{FP}{FP + TN} \quad (16)$$

The overall discriminative ability of the classifier was quantified using the **AUC**, which corresponds to the area beneath the ROC curve. AUC values range from 0.5, indicating random classification, to 1.0, indicating perfect discrimination, with higher values reflecting stronger predictive performance.

Lexical feature distributions across authors

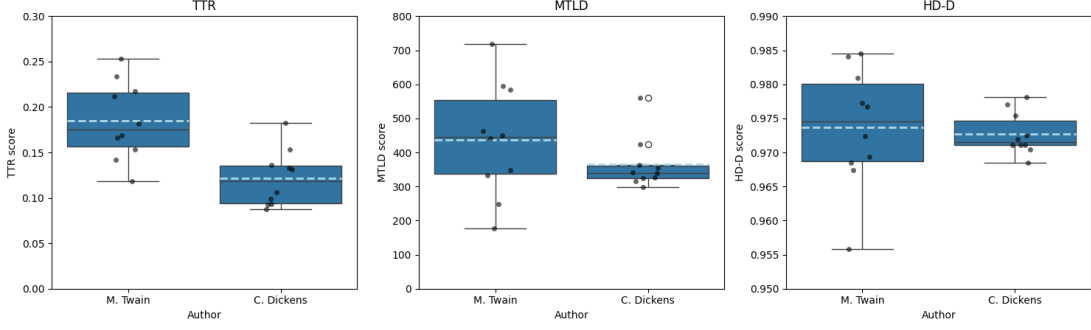


Fig. 3: Lexical feature distributions across authors.

e) Regression Coefficients: Finally, **Logistic Regression coefficients** were examined to assess the relative contribution of each stylistic feature to the classification process. The magnitude of a coefficient reflects the strength of a feature’s influence on the predicted outcome, while its sign indicates the direction of the relationship.

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n \quad (17)$$

where p denotes the probability of belonging to a given class, X_i the predictor variables, and β_i the corresponding regression coefficients.

Consequently, larger absolute coefficient values indicate features with greater discriminative importance for authorship attribution.

IV. RESULTS AND FINDINGS

A. Stylometric Feature Analysis

1) *Lexical Features:* Lexical variation was examined using three complementary measures of lexical diversity: TTR, MTLD, and HD-D. Together, these metrics provide an overview of vocabulary richness and the consistency of lexical variation across texts.

Regarding **TTR** and **MTLD**, Twain consistently exhibits higher levels of lexical diversity than Dickens (Figures 3 and 4). In addition, Twain’s distributions appear considerably more dispersed, indicating greater variability across individual works, whereas Dickens demonstrates a more concentrated and consistent lexical profile.

By contrast, **HD-D** reveals only limited differentiation between the two authors. Both the descriptive statistics and the visualisations show substantial overlap, suggesting broadly comparable levels of lexical diversity when measured using a text-length-independent metric. (Figures 3 and 4).

Overall, the lexical measures indicate a tendency for Twain to employ a more varied vocabulary, while Dickens exhibits greater consistency across works. Nevertheless,

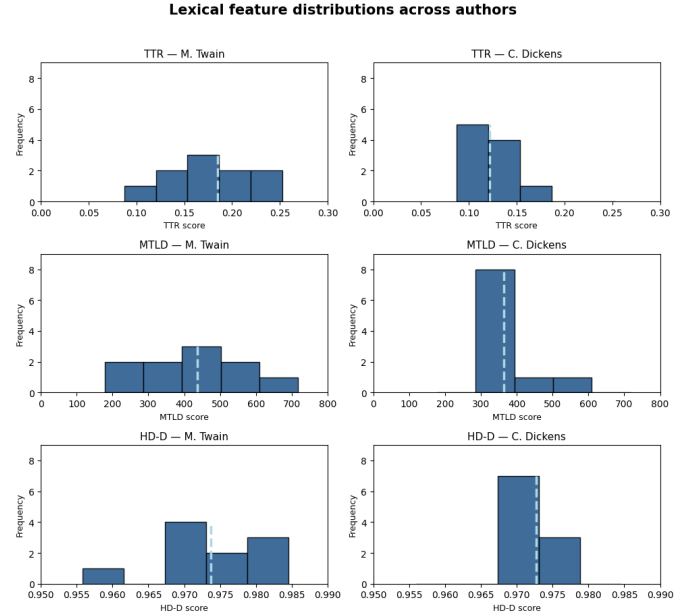


Fig. 4: Lexical feature distributions across authors.

the overlap observed in HD-D suggests that lexical characteristics alone may not provide a complete basis for author discrimination.

2) *Syntactic Features:* Syntactic structures were evaluated using ASL and the PVR.

Regarding **ASL**, Dickens exhibits slightly longer sentences than Twain, though Twain’s distribution is more dispersed, indicating greater variability in sentence length (Figure 5). By contrast, Dickens maintains a more consistent syntactic profile across the corpus.

A similar pattern emerges for the **PVR** (Figure 5). Although Dickens presents a marginally higher usage, the distributions largely overlap. Notably, Twain displays significantly greater variation, whereas Dickens exhibits a highly stable rate of passive constructions.

Overall, these indicators suggest moderate differences between the two authors. Dickens tends to favour longer

Syntactic feature distributions across authors

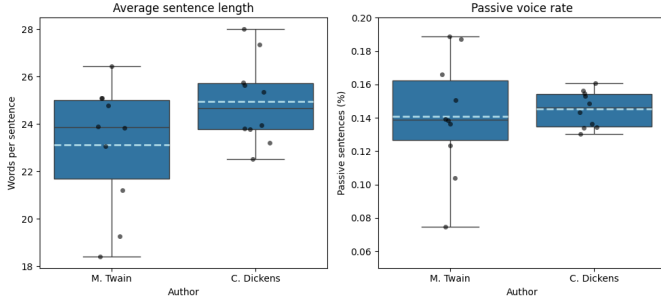


Fig. 5: Syntactic feature distributions across authors.

sentences and passive structures, while Twain is characterised by greater stylistic variation between his works.

3) *Grammatical Features*: Grammatical profiles were examined by analysing the distribution of core POS categories across the corpora.

The overall analysis revealed only limited macro-level differences between the two authors, with both corpora exhibiting highly similar grammatical structures (Figures 6 and 7). Specifically, Dickens employs slightly higher proportions of *nouns* and *pronouns*, whereas Twain presents marginally higher frequencies of *verbs* and *adverbs*. *Adjective* usage remains virtually identical.

Overall, the substantial overlap across all POS categories suggests that grammatical distributions are highly stable and provide limited discriminatory power for authorship attribution.

4) *Affective Features*: Affective characteristics were examined through **VADER sentiment analysis**, utilising *negative*, *neutral*, *positive*, and *compound* scores to quantify the emotional profile of each text.

The results indicate considerable similarity in *negative*, *neutral*, and *positive* sentiment, with the distributions exhibiting substantial overlap (Figures 8 and 9). The most notable differentiation emerges in the *compound* sentiment score, where Dickens displays consistently higher values and greater variability. Conversely, Twain’s scores are lower and more concentrated, suggesting a less polarised emotional tone.

Overall, while most sentiment dimensions reveal broadly comparable affective profiles, the *compound* score provides clear evidence of differentiation, highlighting a more positive emotional orientation in Dickens’ works.

B. Modelling Results

1) *Correlation Analysis*: First, a **Pearson correlation matrix** was examined to explore relationships among the extracted stylistic features and identify potential patterns of feature dependency (Figure 10).

Overall, the heatmap reveals several moderate to strong associations between features belonging to similar stylistic dimensions. Within the lexical and grammatical character-

istics, MTLD and HD-D exhibit a strong positive correlation, indicating that both metrics capture related aspects of lexical diversity. Similarly, *noun* percentage displays a strong positive association with *adjective* percentage, whereas *verb* percentage is negatively correlated with both variables, reflecting expected grammatical trade-offs in word-class usage. Across dimensions, the PVR demonstrates moderate positive correlations with both lexical diversity measures and sentence length, suggesting that these stylistic characteristics frequently co-occur.

With regard to affective features, *positive* sentiment is strongly and positively associated with the *compound* score, whereas *negative* sentiment correlates negatively with both *neutral* and *compound* metrics, aligning with the underlying design of VADER. Crucially, these strong correlations explain why the *compound* score serves as the primary differentiator between the authors: since individual *positive* and *negative* sentiment distributions largely overlap across both corpora, it is the combined interaction of these metrics-captured by the *compound* score that highlights Dickens’ more positive and varied emotional orientation compared to Twain’s concentrated tone.

In sum, these interconnected stylistic patterns indicate that while certain variables co-vary, most features retain sufficient independence to justify their inclusion in the subsequent PCA and classification stages.

2) *PCA*: **PCA** was performed to investigate the underlying structure of the stylistic feature space and to identify the dominant dimensions of stylistic variation. The results were examined through explained variance metrics, PCA projections, and component loadings (Figures 11, 12, 13, 14, and 15).

The **EVR** Analysis indicates that a substantial proportion of the total information is concentrated within the first few principal components. More specifically, PC1 accounts for 32.3% of the total variance, PC2 for 23.7%, and PC3 for 17.5%, jointly explaining approximately 73.5% of the overall variability (Figures 11 and 12). This finding suggests that the majority of stylistic variation within the corpus can be represented by a substantially reduced three-dimensional feature space.

Furthermore, the PCA projections reveal partial separation between the two authors (Figure 13). While PC1 primarily captures general stylistic variability across the corpus, the clearest author-related differentiation is observed along PC2, where the two groups tend to occupy distinct regions of the feature space. PC3 contributes additional discriminatory information, further enhancing author separation.

The **component loadings** and biplot further clarify the variables driving these patterns (Figures 14 and 15). PC1 is mainly associated with TTR and *verb*, *pronoun*, and *adverb* proportions, and opposed by MTLD, HD-D, *noun* percentage, passive syntax, and sentence length. In contrast, PC2 is driven primarily by lexical diversity measures, passive constructions, and sentiment-related fea-

Grammatical feature distributions across authors

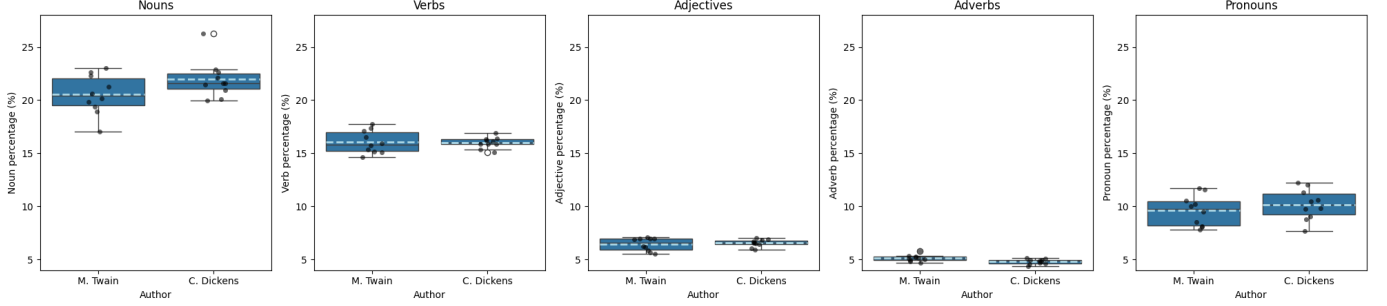


Fig. 6: Grammatical feature distributions across authors.

Grammatical feature composition across authors

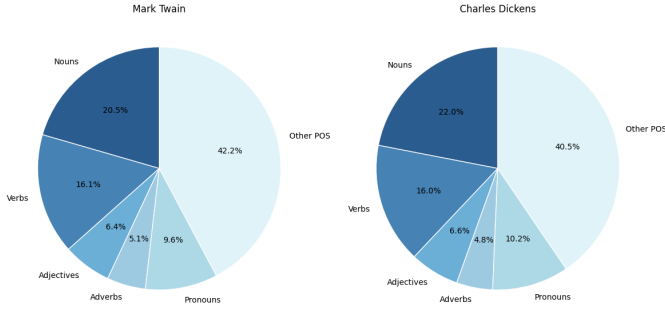


Fig. 7: Grammatical feature composition across authors.

tures, making it the most informative component for author discrimination. Finally, PC3 is characterised mainly by *positive* sentiment, *compound* sentiment, and *pronoun* usage, providing complementary stylistic information.

Overall, the PCA results indicate that the most discriminative dimensions of authorial style are associated with lexical diversity, passive constructions, and sentiment-related characteristics, which dominate PC2 and partially PC3. By contrast, variation in grammatical usage contributes primarily to broader stylistic variability captured by PC1 and appears less effective for author separation. Consequently, differences in vocabulary richness, syntactic complexity, and affective expression emerge as the principal factors distinguishing the literary styles of Twain and Dickens.

3) *Binomial Logistic Regression Analysis*: To evaluate the effectiveness of the extracted stylometric features for authorship attribution, a **Binomial Logistic Regression classifier** was trained and evaluated using Stratified 5-Fold Cross Validation.

Using a decision threshold of 0.50, the classifier correctly identified sixteen of the twenty literary works, yielding four misclassifications (Figure 16). More specifically, eight works by Twain and eight works by Dickens were correctly classified, while two texts from each author were assigned to the opposite class.

TABLE II: Classification Performance Metrics

Metric	Value (%)
Precision	80.0
Recall	80.0
Accuracy	80.0
F1-score	80.0

The resulting **performance metrics** indicate balanced classification behaviour (Table II). Precision of 80% suggests that most predicted author labels were correct, while Recall of 80% indicates that the majority of texts belonging to each author were successfully identified. Similarly, the F1-score of 80% reflects a balanced trade-off between Precision and Recall. Overall, an Accuracy of 80% demonstrates that the proposed stylometric framework is capable of correctly attributing authorship for the majority of texts within the corpus.

Additional insight into model discrimination is provided by the **ROC curve** (Figure 17). The curve remains consistently above the random-classifier baseline and reaches a TPR of approximately 0.80 at virtually zero FPR, indicating strong discrimination at favourable classification thresholds.

The corresponding **AUC value** of 0.85 further confirms the effectiveness of the classifier. This result implies that, when presented with a randomly selected pair of texts from different authors, the model has an approximately 85% probability of assigning a higher classification score to the correct author. Although some classification loss remains evident through the observed misclassifications, such performance is particularly encouraging given the relatively small corpus size employed in the present study.

Ultimately, the **regression coefficients** provide insight into the features contributing most strongly to author discrimination (Figure 18). The largest positive coefficients are associated with *adverb* percentage, MTLTD, *negative* sentiment, and TTR, indicating that these variables increase the probability of assignment to one author class. Conversely, *noun* percentage, *compound* sentiment,

Affective feature distributions across authors

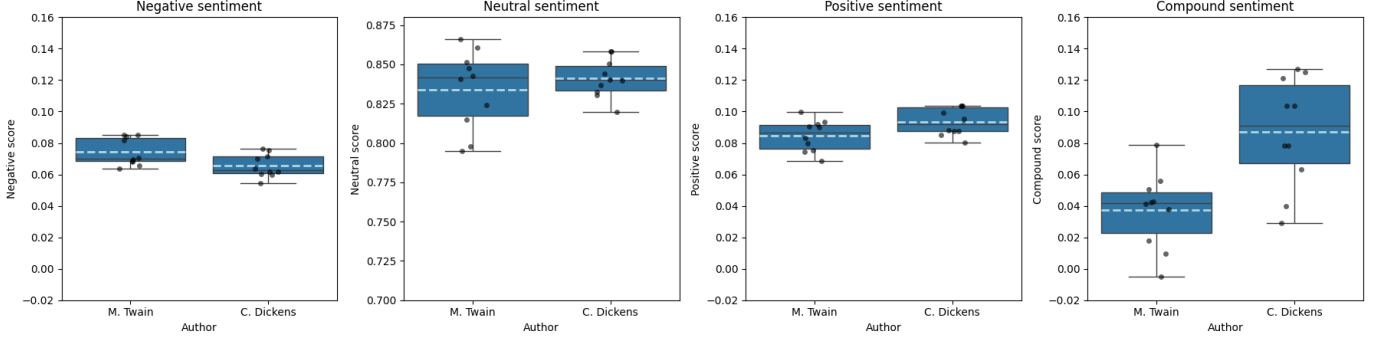


Fig. 8: Affective feature distributions across authors.

Affective feature distributions across authors

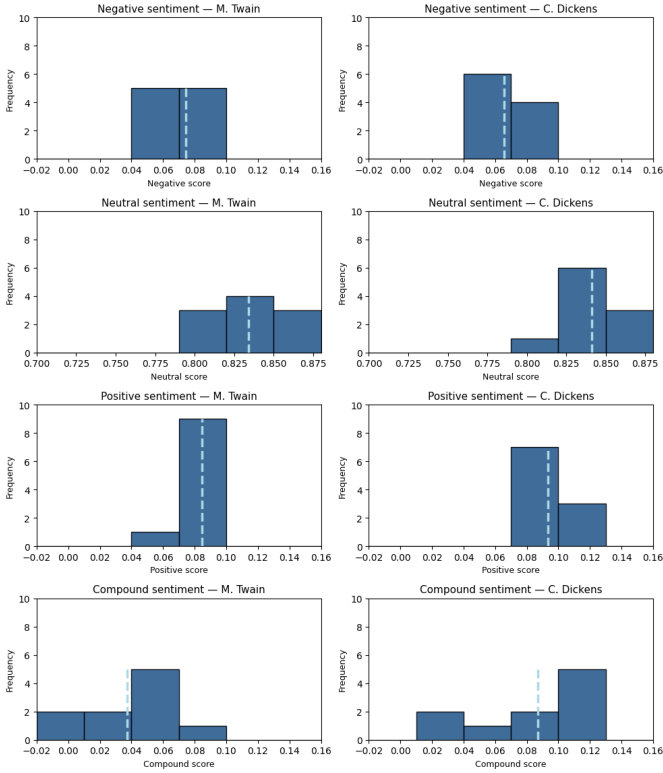


Fig. 9: Affective feature distributions across authors.

positive sentiment, and *pronoun* percentage exhibit the strongest negative coefficients and therefore contribute most strongly toward the alternative class.

Several of these findings are consistent with the PCA results. In particular, lexical diversity measures (MTLD and TTR) and sentiment-related variables emerged as influential contributors both within the principal component structure and within the classification model. ASL also exhibits a moderate coefficient, suggesting that syntactic complexity contributes to author discrimination, albeit

Pearson correlation matrix of stylistic features

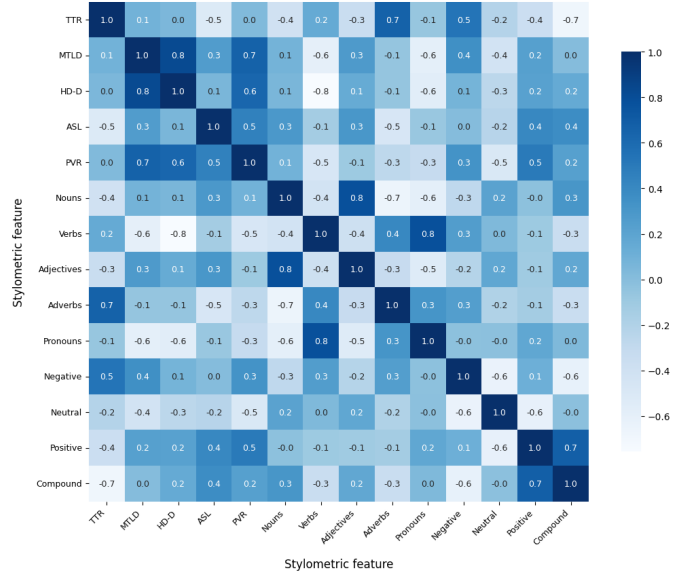


Fig. 10: Pearson correlation matrix of stylistic features.

to a lesser extent than lexical and affective features. By contrast, features such as *verb* percentage, *adjective* percentage, PVR, and HD-D exhibit relatively small coefficients, suggesting a more limited contribution to authorship discrimination despite contributing to broader stylistic variation.

Overall, the coefficient patterns indicate that lexical diversity and affective characteristics constitute the most discriminative dimensions of authorial style within the corpus, with syntax structures providing additional discriminatory information. In contrast, several grammatical features provide comparatively little classification value, reinforcing earlier findings from the PCA and grammatical analyses.

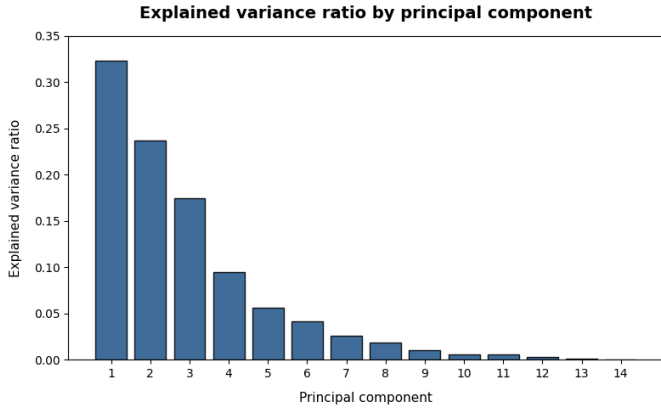


Fig. 11: Explained Variance Ratio (EVR) by principal component.

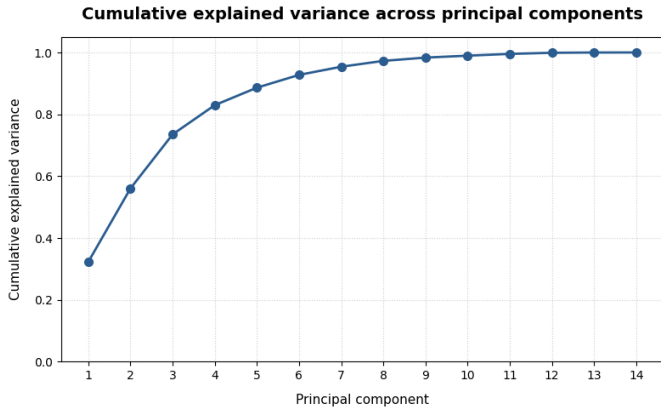


Fig. 12: Cumulative Explained Variance Ratio (EVR) across principal components.

V. INTERPRETATION AND DISCUSSION

A. What Defines Mark Twain’s Literary Style?

The results indicate that Twain’s literary style is primarily characterised by greater vocabulary richness and stylistic variability. Across both TTR and MTLT, Twain consistently exhibits higher values and wider distributions, confirming a less uniform lexicon across his works. This pattern is further reinforced by the PCA results, where lexical diversity measures contribute substantially to the dimensions separating the two authors.

Syntactically, Twain employs slightly shorter sentences and displays greater variation in sentence length and passive voice usage, suggesting a less structurally constrained writing style than Dickens. While grammatical distributions provide limited discriminatory information, Twain exhibits marginally higher frequencies of *verbs* and *adverbs*, reflecting a more dynamic linguistic profile.

Regarding affective features, Twain’s works generally display lower *compound* sentiment scores and a more *neutral* emotional profile. In the PCA space, his texts align closely with PC2, a component driven by lexical diversity.

Combined, these findings demonstrate that Twain’s style is defined by vocabulary breadth and linguistic flexibility rather than stable grammatical and sentiment preferences.

B. What Defines Charles Dickens’ Literary Style?

Dickens exhibits a more consistent stylistic profile across the corpus. Lexical diversity measures are generally lower than Twain’s, with considerably tighter distributions indicating greater consistency in vocabulary usage.

Syntactically, Dickens employs slightly longer sentences and a more stable rate of passive constructions. Although passive voice usage differs only modestly between the authors, its low variability suggests a uniform structural style. Grammatically, *noun* and *pronoun* frequencies are marginally higher, though overall POS distributions contribute little to author differentiation.

The most distinctive affective characteristic of Dickens’ writing is the consistently higher *compound* sentiment score, which suggests a more *positive* emotional tone. In the PCA space, his texts align with the negative side of PC2, a dimension driven by reduced lexical diversity. Consequently, Dickens’ style is characterised by lexical consistency and syntactic regularity, driven by distinct grammatical and sentiment preferences.

C. Which Stylometric Features Contribute Most to Author Differentiation?

The combined evidence from the PCA loadings and Logistic Regression coefficients suggests that not all stylometric features contribute equally to author discrimination. Lexical diversity measures, particularly TTR and MTLT, consistently emerged among the strongest contributors across both analyses, indicating that vocabulary richness constitutes a major source of stylistic differentiation between the two authors.

Affective characteristics also proved highly informative. Sentiment-related variables contributed substantially to the separation observed in the PCA projections and received relatively large weights within the classification model. In contrast, grammatical features, such as POS distributions, exhibited limited discriminatory power, despite contributing to broader stylistic variation within the corpus.

Sentence length and passive voice usage provided additional evidence of stylistic differentiation, although their contribution was generally weaker than that of lexical and affective features. Overall, the findings suggest that authorial style is determined primarily by differences in vocabulary usage, syntactic complexity, and emotional expression rather than by grammatical composition alone.

D. How Effective is Stylometric Authorship Attribution?

The findings indicate that stylometric analysis can successfully distinguish between the literary styles of Twain and Dickens. First, the PCA projections revealed meaningful separation between the two authors, particularly along

Principal component projections of literary works

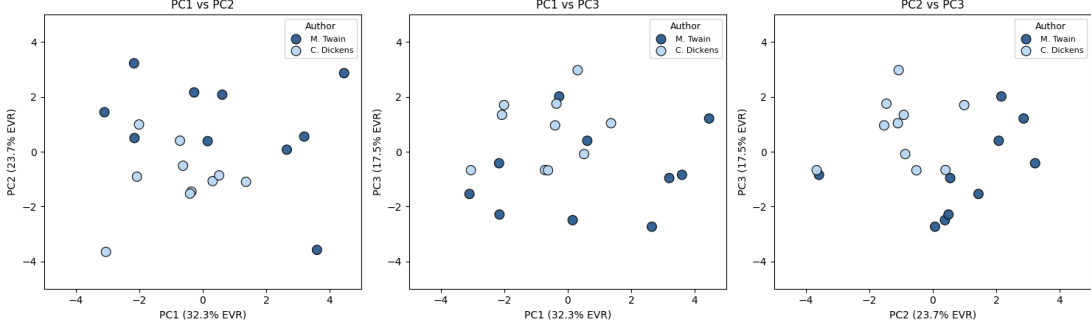


Fig. 13: Principal Component Analysis (PCA) projections of literary works.

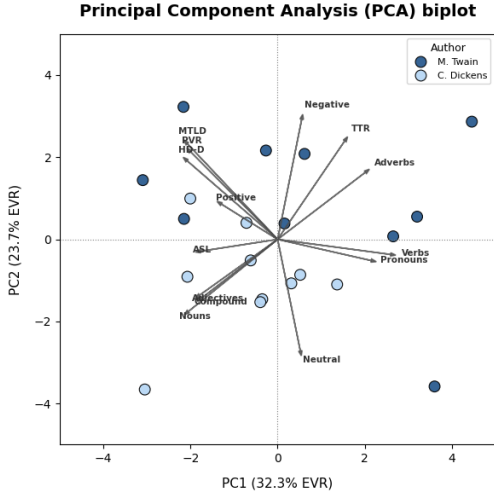


Fig. 14: Principal Component Analysis (PCA) biplot.

the PC2, suggesting the presence of identifiable stylistic structure within the extracted features.

This observation is further supported by the classification results. The Logistic Regression model achieved an overall accuracy of 80%, while the ROC analysis yielded an AUC value of 0.85, indicating strong discriminatory performance. Although a small number of texts were misclassified, the classifier consistently performed well across all evaluation metrics.

Taken together, these findings demonstrate that stylistic features contain sufficient information to support automated authorship attribution within the examined corpus. More broadly, the results highlight the potential of computational approaches to identify and quantify meaningful stylistic differences between authors.

E. Methodological Limitations

In conclusion, several limitations should be acknowledged when interpreting the findings of the present study. First, the corpus consists of only twenty literary works, which restricts statistical generalisability and may limit

the stability of the derived models. Second, the present analysis is confined to English-language texts and therefore cannot be assumed to generalise across languages or literary traditions.

In addition, certain stylistic features were extracted using heuristic or rule-based procedures. In particular, passive voice detection relied on predefined grammatical patterns and may not capture all syntactic variations occurring in literary texts. Similarly, sentiment analysis was based on the VADER lexicon, which was originally designed for general-purpose language and may be less effective in capturing irony, figurative language, or context-dependent meanings commonly found in literature.

The selection of only two authors also introduces a degree of author-selection bias, as the observed patterns may partially reflect the specific characteristics of Twain and Dickens rather than universal dimensions of literary style. Finally, computational stylistic analysis should not be viewed as a replacement for traditional literary scholarship. Rather, it constitutes a complementary methodological framework that can enrich, support, and extend insights derived from close reading and qualitative interpretation.

VI. CONCLUSION

The present study demonstrated that stylistic analysis can effectively distinguish between the literary styles of Twain and Dickens. The findings revealed that lexical diversity, sentiment-related features, and sentence length provided the strongest discriminatory information, whereas grammatical distributions contributed comparatively little. PCA successfully identified meaningful stylistic dimensions and author-related patterns, while Logistic Regression achieved an overall classification accuracy of 80% and an AUC value of 0.85, supporting the effectiveness of the proposed authorship attribution framework.

Ultimately, this study contributes to the growing field of computational stylistics and, by its extension, the DH, by demonstrating the value of combining stylistic features with statistical and machine learning techniques

Principal component loadings of stylistic features

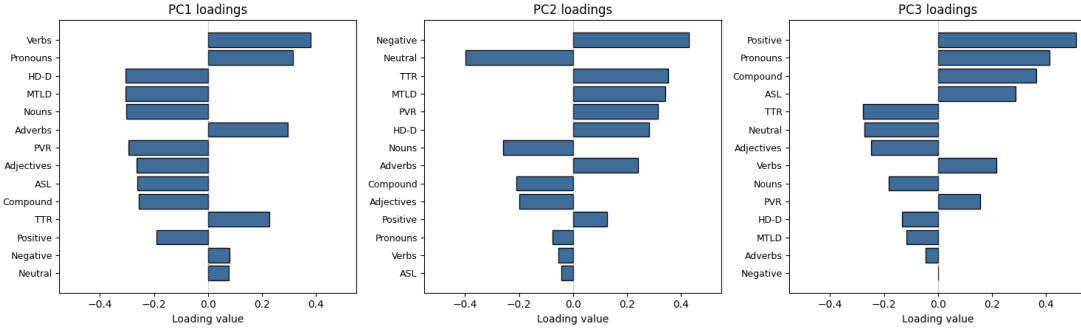


Fig. 15: Principal component loadings of stylistic features.

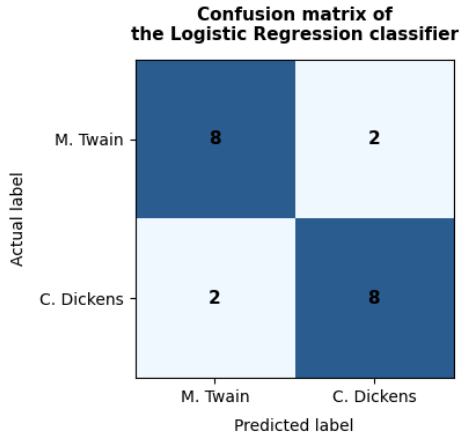


Fig. 16: Confusion matrix of the Logistic Regression classifier.

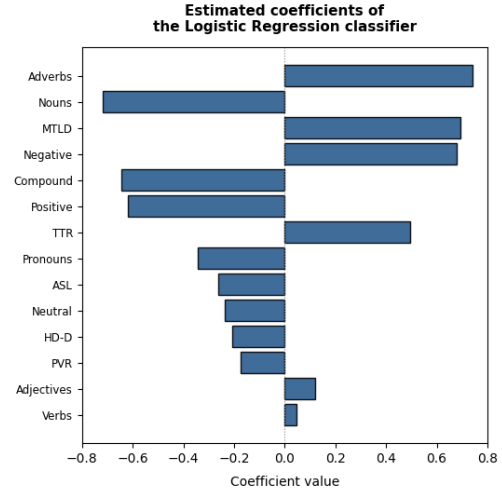


Fig. 18: Estimated coefficients of the Logistic Regression classifier.

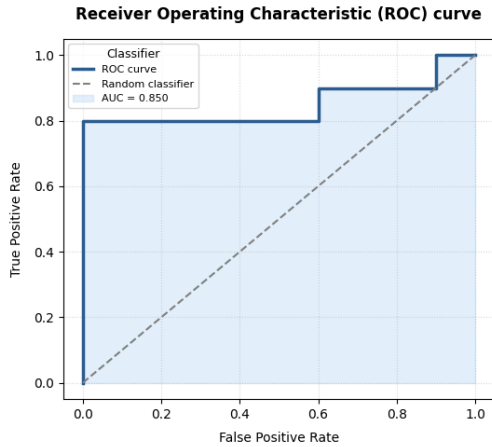


Fig. 17: Receiver Operating Characteristic (ROC) curve.

for literary analysis. Future research could incorporate larger corpora, additional authors, more advanced linguistic features, and alternative classification algorithms to further improve authorship attribution performance.

ACKNOWLEDGEMENTS

The author would like to thank the instructor of the *Data Analysis for Humanities with Python* course for providing methodological guidance. The author also gratefully acknowledges the *Union of Greek Shipowners* and the *Georgios and Andrianí Lordos Foundation* for the financial support provided through the award of scholarships that enabled the completion of his postgraduate studies.

REFERENCES

- [1] J. F. Burrows, "Delta: A Measure of Stylistic Difference and a Guide to Likely Authorship," *Literary and Linguistic Computing*, vol. 17, no. 3, pp. 267–287, 2002.
- [2] Encyclopaedia Britannica, "Mark Twain," Available: <https://www.britannica.com/biography/Mark-Twain> [Accessed: 02 June 2026].
- [3] Encyclopaedia Britannica, "Charles Dickens," Available: <https://www.britannica.com/biography/Charles-Dickens-British-novelist> [Accessed: 02 June 2026].
- [4] Project Gutenberg. Available: <https://www.gutenberg.org> [Accessed: 02 June 2026].
- [5] D. Jurafsky and J. H. Martin, *Speech and Language Processing*, 3rd ed. Online draft manuscript, 2024.

- [6] P. M. McCarthy and S. Jarvis, “MTLD, vocd-D, and HD-D: A Validation Study of Sophisticated Approaches to Lexical Diversity Assessment,” *Behavior Research Methods*, vol. 42, no. 2, pp. 381–392, 2010.
- [7] C. J. Hutto and E. Gilbert, “VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text,” in *Proceedings of the Eighth International AAAI Conference on Weblogs and Social Media (ICWSM)*, pp. 216–225, 2014.
- [8] H. Abdi and L. J. Williams, “Principal Component Analysis,” *Wiley Interdisciplinary Reviews: Computational Statistics*, vol. 2, no. 4, pp. 433–459, 2010.
- [9] C. Y. J. Peng, K. L. Lee, and G. M. Ingersoll, “An Introduction to Logistic Regression Analysis and Reporting,” *The Journal of Educational Research*, vol. 96, no. 1, pp. 3–14, 2002.